
INTERRATER RELIABILITY AND AGREEMENT

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The rating scale is a ubiquitous measuring instrument, enjoying widespread usage in popular culture, the physical, biological and social sciences, and the humanities. Rating scales occur in a great variety of forms; most use numbers to reflect gradations in the attribute being judged, but some use pictures or other symbols (i.e., thumbs up vs. thumbs down, one to five stars, one to five mice, or one to five cows). Regardless of the cosmetic differences in appearance, rating scales typically require a rater, judge, or observer (we use the terms interchangeably in this chapter) to evaluate some characteristic of an object and assign it to some point on the rating scale. The object to be rated can be a person (e.g., an employee or counselor), a thing (a book, music CD, or movie), a process (e.g., types of counseling or recovery from anesthesia), an outcome (e.g., recovery of function or assessment of apraxia), an idea (e.g., quality of life or contribution to society) or a datum (e.g., time since last update or occurrence of target behavior).

Despite the widespread popularity of rating scales, few applied researchers seem to understand the design issues unique to the collection of ratings nor the proper procedures for evaluating ratings. The quality of ratings can be critical to the success of your research, however, because

the effect size and power of your analyses are contingent on maintaining measurement error within acceptable limits. Failure to maintain the reliability of your ratings data (and, possibly, the level of agreement among your raters) within an acceptable level may make it impossible for your research to yield noteworthy effect sizes. Yet Thompson (1994) and Fabrigar, Wegener, MacCallum, and Strahan (1999) estimate that reliability is given insufficient attention in 40 to 50% of the published literature, and Pedhazur and Schmelkin (1991) refer to the lack of attention given to measurement as the Achilles heel of research in the social and biological science.

This problem is even more serious when working with rating scales because the datum recorded on a rating scale is the subjective judgment of an individual (i.e., the rater). It is always necessary to establish the reliability of a set of ratings to demonstrate that the variance in the ratings is due to a systematic rather than a random ordering of the objects. Knowledge of the interrater reliability of a set of ratings is crucial in evaluating the validity and generalizability of the results, yet the failure to report the interrater reliability of ratings is not uncommon. Another common practice is to report the interrater reliability as having been determined in a previous investigation. Those who do so apparently do not grasp the critical point that reliability is not a property of the rating scale, but a property of a specific set of data that result from a specific use of the scale. Knowledge of the interrater agreement of a set of ratings also may be important, yet the interrater agreement of ratings seldom is reported. These practices are unacceptable.

In this chapter we explain the critical distinction between interrater reliability and agreement, a distinction that is given little attention in other areas of measurement (see Dawis, chapter 3, this volume). We review design issues such as the type of replication and the level of measurement (see Tinsley & Brown, chapter 1, this volume and Dawis, chapter 3, this volume) that researchers must consider in determining how to collect and analyze ratings data. However, the primary purpose of this chapter is to provide guidelines for selecting the appropriate statistical procedure for analyzing ratings data. Determining the proper procedure for calculating interrater reliability and interrater agreement requires a consideration of the level of measurement achieved by the rating scale and the intended use of the ratings. We also consider briefly the desirability of using weighting schemes for transforming nominal scale ratings to ordinal scale ratings.

I. AGREEMENT VERSUS RELIABILITY

One critical distinction that applied researchers must understand to properly evaluate the psychometric quality of ratings data is that between interrater reliability and interrater agreement. This distinction is analogous

to coming to a T-shaped intersection while on a Sunday drive; it controls the direction your analysis will take. Making the wrong decision most likely means that your data will not be analyzed properly no matter what decisions you make later.

Procedures for analyzing interrater reliability were suggested as early as the 1930s (e.g., Burt, 1936; Hoyt, 1941), but formal introduction of the intraclass correlation (R) did not occur for another decade (e.g., Bartke, 1966; Ebel, 1951; Guilford, 1954; Lindquist, 1953). Widespread confusion regarding the difference between interrater reliability and agreement continued until the mid-1970s, however, with most researchers using those terms incorrectly as synonyms. For example, K. Lu (1971) proposed a measure of interrater agreement that is conceptually identical to a measure of interrater reliability proposed earlier by Finn (1970). Recognition that interrater reliability indices and interrater agreement indices provide fundamentally different information about a set of ratings has greatly improved since we clarified their quintessential difference (Tinsley & Weiss, 1975). Sadly, however, failures to understand this distinction and to use interrater reliability and interrater agreement indices properly still are not uncommon. A quick scan of the literature published in medicine and psychology since 1997 using the Medline database revealed over 30 instances in which interrater agreement indices were represented as measuring interrater reliability. The consequence of these errors is wastefulness.

One noteworthy illustration of the problems that can occur when researchers confuse interrater reliability and interrater agreement is James, Demaree, and Wolf's (1984) proposal of an index of interrater reliability (r_{wg}) for use when a single stimulus has been rated. Schmidt and Hunter (1989) criticized Kozlowski and Hults's (1987) use of this index on the grounds that, "it is not conceptually possible within standard measurement theory to compute a legitimate estimate of the interrater reliability coefficient when raters rate only one stimulus" (p. 370). Schmidt and Hunter (1989) expressed concern that r_{wg} , "may have no meaningful interpretation" (p. 369). The issue was resolved only after Kozlowski and Hattrup (1992) clarified that r_{wg} is actually a measure of interrater agreement, a point that James, Demaree, and Wolf (1993) subsequently acknowledged.

In classical psychometric theory, reliability is defined as the ratio of true score variance to total variance (Crocker & Algina, 1986). In the typical measurement situation in which internal consistency reliability is assessed, a sample of persons complete an instrument consisting of multiple items at a single point in time. Because the status of each person is measured only once, and it is assumed that all of the items are measuring the same construct, any variance in the scores on the instrument should be the result of differences among the persons being tested. Therefore, the variance among persons is interpreted as true score variance and any nonsystematic variance is interpreted as error variance.

Essentially the same situation exists when a rating scale is used, except that the multiple raters take the place of the multiple items. The attribute on which the objects are rated is assumed to be constant at the time of measurement, all of the raters are assumed to be rating the same object, and the variability in the ratings should be the result of differences among the rated objects. Therefore, the variance attributable to differences among the rated objects is interpreted as true score variance, and any additional variance is interpreted as error variance.

Interrater reliability, therefore, provides an indication of the extent to which the variance in the ratings is attributable to differences among the objects rated. As with other forms of reliability, it is not necessary that the rated objects be assigned exactly the same rating for interrater reliability to be high. This is because, just like internal consistency reliability, interrater reliability is calculated using correlational (or analysis of variance [ANOVA]) indices. To understand the significance of that, consider that the pattern of ratings a judge assigns to a set of objects can be thought of as a profile of scores, just as an individual's scores on the 10 clinical scales of the Minnesota Multiphasic Personality Inventory (MMPI) form a profile of scores. Two profiles can differ in level (i.e., the overall mean score of one person is higher than the overall mean score of the second person), scatter (i.e., the two persons' scores differ in the extent to which they vary around the overall mean), and shape (i.e., the two persons' scores have the same overall mean and variability, but their high and low scores occur on different scales). Correlational indices of the degree of association between two profiles are insensitive to differences in level and scatter because they standardize the scores by setting the mean of each set to zero and the standard deviation of each set to 1.0. Therefore, Interrater reliability indicates the degree to which the ratings of different judges are the same when expressed as deviations from their means. In practice, a high interrater reliability means that the relation of one rated object to other rated objects is the same across judges, even though the absolute numbers used to express this relation may differ from judge to judge. Interrater reliability is sensitive only to the relative ordering of the rated objects.

In contrast, interrater agreement represents the extent to which the different judges tend to assign exactly the same rating to each object. As such, interrater agreement indices are sensitive to differences in the level, scatter, or shape of the profile of ratings assigned by the judges. When judgments are made using a numerical scale, a high interrater agreement signifies that the judges assigned precisely the same numerical values when rating the same person.

Table 4.1 shows hypothetical data (assuming interval-level measurement) in which three judges have rated 10 job applicants on a 10-point applicant rating scale. Case 1 illustrates a situation in which all three judges assign exactly the same ratings to each of the 10 applicants. These ratings

TABLE 4.1 Hypothetical Ratings of Job Applicants Illustrating Different Levels of Interrater Reliability and Interrater Agreement for Interval-Scaled Data

Candidate	Case 1: High interrater reliability and high interrater agreement			Case 2: High interrater reliability and low interrater agreement			Case 3: Low interrater reliability and high interrater agreement		
	Rater			Rater			Rater		
	1	2	3	1	2	3	1	2	3
A	1	1	1	1	3	6	5	6	5
B	2	2	2	1	3	6	5	4	4
C	3	3	3	2	4	7	6	4	6
D	4	4	4	2	4	7	4	5	6
E	5	5	5	3	5	8	5	4	4
F	6	6	6	3	5	8	6	6	5
G	7	7	7	4	6	9	4	4	5
H	8	8	8	4	6	9	5	5	4
I	9	9	9	5	7	10	4	5	3
J	10	10	10	5	7	10	6	6	6
Descriptive statistics									
Mean	5.5	5.5	5.5	3.0	5.0	8.0	5.0	5.1	4.7
SD	3.0	3.0	3.0	1.5	1.5	1.5	.8	.9	.8
Interrater reliability									
R_{di}		1.00			1.00			.38	
R_{gi}		1.00			.02			.40	
R_{dc}		1.00			1.00			.65	
R_{gc}		1.00			.06			.67	
r_i		1.00			.23			.94	
r_c		1.00			.47			.98	
Interrater agreement									
T_0		1.00			.00			.00	
T_1		1.00			.00			.67	
T_2		1.00			.00			1.00	

have both high interrater agreement and high interrater reliability, and the distinction between these indices is blurred.

Case 2 shows ratings having high interrater reliability but low interrater agreement. These data have high reliability because the ratings assigned to the job applicants by the three judges are the same when expressed as deviations from their overall mean. You can verify this for yourself by subtracting the judge's mean from each rating. For example, the ratings given the first candidate by judge #1 (i.e., $1 - 3 = -2$), Judge #2 (i.e., $3 - 5 = -2$), and Judge #3 (i.e., $6 - 8 = -2$) are identical when expressed as deviations from each judge's overall mean rating.

Despite the perfect interrater reliability of the ratings in Case 2, these ratings have low interrater agreement because no two raters gave the same rating to any job applicant. We explained that agreement indices are sensitive to differences in the level (i.e., mean), scatter (i.e., variability), and shape of the ratings. In Case 2 the pattern of scores and the variability of the ratings are equal across judges, but you can see the cause of the low agreement among the raters reflected in their mean ratings of 3.0, 5.0, and 8.0, respectively.

Given the similarity of the pattern and variability of the judges' ratings, you might be tempted to dismiss the differences in the mean level of the ratings. To understand the importance of this difference, pretend for a moment that you are applying for a job and could pick one of these judges to evaluate your suitability for employment. Would you care which judge rated you? Of course you would; we all would prefer to be evaluated by rater #3 because it seems likely that the ratings assigned by the three judges will translate into meaningfully different statements about the acceptability of these job applicants. In other situations (e.g., when rating degree of disability or carotid artery plaque morphology) the specific interpretations attached to the different levels of the rating scale may be quite important.

A critical distinction to understand, therefore, is that interrater reliability is a satisfactory index of the quality of the ratings when you are interested only in the relative ordering of the rated objects (i.e., a norm-referenced interpretation). However, interrater reliability indices are insensitive to differences in the mean and variability of the ratings given the rated objects. Whenever the absolute value of the ratings or the meaning of the ratings as defined by the points on the scale is of concern (i.e., criterion referenced interpretation), the interrater agreement index provides critical information about the ratings.

The interrater agreement of the ratings is always critical when judgments about the rated objects are based on information from different sets of judges. Suppose, for example, that you actually have 99 applicants for 10 positions. Two scenarios typically occur in situations such as this. In Scenario One each candidate is rated by only one judge. If each judge rated 33 of the applicants, none of the candidates interviewed by judge 1 or judge 2 would be offered a position. In Scenario Two each candidate is rated by two judges in an effort to provide a broader perspective on each candidate. In this scenario the candidates rated by Judges #1 and #2 and the candidates rated by Judges #1 and #3 would not be offered positions. Only those candidates interviewed by Judges #2 and #3 would have a chance at employment. Although the fallacy of these scenarios may seem obvious, they describe the situation often faced in applied research. Despite best intentions, the rating task is found to exceed the available resources and Scenario One or Two is adopted as an expedient. In situations such as this it is mandatory that interrater agreement be assessed and reported in addition to interrater reliability.

Finally, Case 3 depicts ratings that have high interrater agreement but low interrater reliability. The high interrater agreement results from the fact that the ratings assigned to each job applicant by the three judges are quite similar. The low interrater reliability is the result of the restricted range of ratings given by the judges. This occurs because reliability is defined as the proportion to true score variance attributable to total variance. Therefore, as total variance (the denominator in the equation) approaches zero, reliability approaches zero. Ratings such as these may signify that the job applicants are homogeneous on the trait of interest, or they could indicate that the judges did not use the rating scale properly. This issue can be investigated by further study of the same subjects or by having the judges rate a sample of subjects known to be heterogeneous on the trait of interest. Regardless of the cause, however, interrater reliability will be low when the variability of the ratings is low, as often occurs in applied research.

In summary, interrater reliability and interrater agreement refer to two important and meaningfully different attributes of a set of ratings. High reliability is no indication that the raters agree in an absolute sense on the degree to which the objects rated possess the characteristic being judged (Case 2). On the other hand, low reliability does not necessarily indicate that the raters are in disagreement (Case 3). Both types of information are necessary to evaluate ratings. When both interrater reliability and agreement are low, a case not illustrated in Table 4.1, the ratings have no validity and should not be used for research or applied purposes.

II. LEVEL OF MEASUREMENT

Tinsley and S. Brown discuss the differences in nominal, ordinal, and interval level measurements in chapter 1 (this volume). That distinction is as important in assessing interrater reliability and interrater agreement as it is in using the other multivariate statistics and model testing procedures covered in the *Handbook*. Interval-level statistics often can be used appropriately on ratings that result from ordinal level measurement, but keep in mind that such applications can result in interrater reliability and interrater agreement statistics that distort the true relations among raters when the assumption of equal intervals is grossly inappropriate.

Nevertheless, the critical distinction for interrater reliability and interrater agreement is that between nominal level measurement and higher levels of measurement. The distinction between interrater reliability and interrater agreement ceases to exist when ratings are made at the nominal level of measurement. With nominal ratings scales the rating categories do not differ quantitatively, so the disagreements in categorization do not differ in their severity. Therefore, the concept of "proportionality of rat-

ings," which is central to interrater reliability, ceases to make sense at the nominal-scale level. Agreement becomes an absolute; ratings are either in agreement or disagreement. As a result, the concept of interrater reliability ceases to make sense for nominal scales. The term interrater agreement is more appropriate, because the attribute being measured is the extent to which the judges assigned the same rating to each rated object.

III. TYPE OF REPLICATION

The concept of reliability involves some type of replication or repetition of the measuring process. In classical psychometric theory the measurements are replicated by having the same persons complete two different measures of the same construct (i.e., parallel-forms reliability), by having the same persons complete the instrument at two points in time (i.e., test-retest reliability), or by having the same persons complete multiple items that are presumed to measure the same construct (i.e., internal consistency reliability). In much the same manner, determination of interrater reliability also requires some form of replication. We are not aware of a parallel-forms analog in assessing interrater reliability, but analogs to the test-retest and internal consistency data collection procedures exist.

One method of replicating a set of ratings is the rate-rerate method, the analog of the test-retest procedure for determining test reliability. In this method the judges rate a group of subjects on some characteristic, then rerate the same subjects on the same characteristic at a later date. This replication has been used recently by Dreessen and Arntz (1998) and Chorpita, Brown, and Barlow (1998).

The rate-rerate method is an inappropriate procedure for demonstrating interrater agreement or interrater reliability for several reasons. First, this measure of reliability provides no information about interrater agreement because it treats the raters as a constant and ignores between-judges variability in the ratings. The rate-rerate method could yield a high rate-rerate reliability for three raters who were in complete disagreement, so long as each rater was consistent across time. Second, the rated objects may change over time on the characteristics being rated. The more accurately the judges amend their previous ratings to reflect this change, the lower the reliability will appear. Third, rate-rerate methods may give spuriously high reliabilities when the between-ratings interval is short and a nonchanging database is used (e.g., behavioral ratings of audio- or videotape excerpts). In this situation the second set of ratings will likely be contaminated by the raters' recall of their previous judgments. Finally, the rate-rerate method requires that the judges rate each subject on two different occasions. This is seldom feasible in the field situations in which much applied research is conducted.

For these reasons we advocate replication of ratings between judges (i.e., have several judges rate the same objects on the same attribute), as illustrated in Table 4.1. This procedure is the analogue of internal consistency reliability with the multiple judges taking the place of the multiple items.

IV. INTERRATER RELIABILITY

Students of classical psychometric theory know that approaches, to the calculation of reliability based on the correlation coefficient (Kuder & Richardson, 1937) and ANOVA (Hoyt, 1941) offer psychometrically different starting points for arriving at the same destination. Hoyt's (1941) reliability and Cronbach's (1951) coefficient alpha reliability are algebraically equivalent, and both produce the same result as the Kuder-Richardson-20 (Kuder & Richardson, 1937) formula when applied to dichotomous data. A similar phenomenon occurred in the development of measures of interrater reliability. Numerous procedures were proposed for calculating interrater reliability prior to Ebel's (1951) now classic work on the intraclass correlation (R ; e.g., Gulliksen, 1950; Horst, 1949; Peters & Van Voorhis, 1940), and additional procedures continue to be recommended (e.g., Finn, 1970), some of which are represented as measures of interrater agreement (e.g., K. Lu, 1971). These procedures yield values that are identical to or approximate a value obtained from one of the forms of R . Although we will describe one situation in which Finn's r is useful, we advocate the use of R in most instances.

Generalizability theory (Cronbach, Gleser, Nanda, & Rajaratnam, 1972; see Marcoulides, chapter 18, this volume) provides a conceptual framework within which to understand the intraclass correlation. Generalizability theory introduced the notion that test scores (and ratings) are influenced by multiple forms of error, and that different measures of reliability can be calculated to assess the influence of these sources of error. Multiple formulations of R are possible for precisely this reason (see Ebel, 1951; Hoyt, in press; Shrout & Fleiss, 1979; Tinsley & Weiss, 1975); in this chapter we discuss four variations that have the greatest applicability in applied research. Generalizability theory provides a structured framework within which investigators can compare the alternative forms of the intraclass correlation in terms of the (a) type of error assessed and conceptual meaning of the reliability index, (b) assumptions underlying the index, (c) relevance of the index to their research objectives, and (d) costs of the index.

Although technically, R requires the assumption of interval-level ratings, we recommend that R be used with ordinal scales that assume interval properties unless the variance in the ratings is severely restricted (as in Case 3). We will discuss the use of Finn's (1970) r in that situation later

in this chapter. R can be interpreted as the proportion of the total variance in the ratings due to variance in the persons being rated (i.e., as the proportion of variance attributable to true score variance). As such, R is consistent with the classical definition of reliability. In actual practice, R can be expected to range from 1.00 to zero, with high values indicating high interrater reliability (i.e., most or all of the variance in the ratings is attributable to differences among the objects rated). Values close to zero indicate a complete lack of reliability (viz. almost all of the variability in the ratings is due to error). Although negative values of R are mathematically possible, they are rarely observed in actual practice. Negative values are most likely to occur when the variance in the ratings is severely restricted, or when a Rater $\times$ Object interaction is present (e.g., Judge 1 rates men high and women low, whereas Judge 2 does the reverse).

Two decisions are necessary to decide on the proper approach to calculating R . Ebel (1951) pointed out that R permits investigators to treat the between-rater variance as error variance or to exclude it from the error term. In order to make this decision, researchers must decide whether differences in the level (i.e., mean) or scatter (i.e., variance) in the ratings of the judges represent error or inconsequential differences. In addition, investigators must decide whether they want to know the average reliability of the individual judge or the reliability of the composite rating of the panel of judges. The product of these two dichotomous decisions yields four versions of R . We will discuss each of these decisions in turn.

A. Between-Raters Variance

1. Interpretation of Between-Raters Variance

Case 2 in Table 4.1 illustrates a situation in which the judges assigned identical rankings to job candidates, but they differed in their average ratings (i.e., the mean ratings were 3.0, 5.0, and 8.0 for judges 1, 2, and 3, respectively). These "level" differences account for all of the difference in the judges ratings. It stands to reason, therefore, that elimination of these differences in the average rating given by the judges (i.e., the between-judges level differences) would yield data having perfect reliability (i.e., $R = 1.00$). Some writers have suggested that this formulation of interrater reliability be interpreted as a measure of interrater agreement (e.g., Suen & Ary, 1989), a point to which we will return shortly. On the other hand, the decision to treat the between-judges variance as error results in an intraclass correlation of .017. Whenever differences are apparent in the mean or variance of the judges' ratings, the decision regarding the interpretation of between-raters variance will influence the interrater reliability of the data. If sizable differences in mean or variance are present, the exclusion of the between-raters variance from the error term will cause the reliability coefficient to be substantially higher than if it were included.

The desirability of removing the interjudge variance from the error term when estimating the interrater reliability depends on the way in which the ratings are to be used (Ebel, 1951). Do not include the between-raters variance in the error term if these differences do not influence the interpretation of the ratings. That will be the case when decisions are based on the mean or sum of the ratings assigned by the set of observers, or on ratings that have been adjusted (e.g., by conversion to ranks or Z scores) to eliminate the between-rater differences. Include the between-raters variance in the error term if (a) decisions are made by comparing objects rated by different judges or different sets of judges, or (b) the investigator wants to generalize the results to other samples of judges using the same scale with a similar sample of subjects.

2. Generalizability of R

Several writers suggest or imply that the ability to generalize the interrater reliability is desirable (Shrout & Fleiss, 1979; Suen & Ary, 1989). We maintain that applied researchers should make every effort to design their research so that between-judges variation can be removed from the error term; as can be seen in Case 2, the cost of treating between-judges variance as error can be great. Excluding between-judges variation from the error term means that the sample of raters used in the computation of R must be regarded as "fixed" (i.e., they represent the population of judges rather than a sample of judges from some population; Burdock, Fleiss, & Hardesty, 1963; Tinsley & Weiss, 1975). The cost of this decision is that the calculated R is a descriptive statistic, applicable to only the data on which it was calculated, and therefore it cannot be generalized to any other set of judges using the same rating scale on a similar sample of persons.

This cost is not as great as it may appear, however, for it is seldom feasible to generalize an interrater reliability coefficient for both conceptual and practical reasons. Conceptually, as we noted earlier, it is now widely agreed that reliability is not a generalizable property of a rating scale, but a descriptive property of a specific set of data that result from a specific use of the scale. Beyond that, researchers would need to satisfy several rather difficult requirements to generalize the calculated R to another sample of judges. First, a random sample of judges is necessary so that the judges can be considered representative of the population of judges to which you wish to generalize. Second, the new judges must use the same rating scale. Third, the judges must be trained to the same level of proficiency as the original judges. Fourth, the objects to be rated must be the original objects or a random sample from the same population as the original objects. Finally, the equation for R must incorporate an estimation of the between-judges variance so that the degree to which R will vary across samples of judges can be estimated.

It seems unlikely that the requirement of a random sample of judges will be satisfied in applied research, and the original investigator has no control over the selection of the judges, training of the judges, or selection of objects to be rated in the subsequent research unless he or she is conducting that research. Therefore, despite your best intentions and your willingness to accept a lower reliability, it seems unlikely that the reliability obtained in your research will generalize to other studies. Ebel (1951) provided formulas for establishing confidence intervals around R , but in most applied research studies the lower limit of the confidence interval falls below zero because of the small number of raters used. This further emphasizes the folly of attempting to generalize interrater reliability coefficients. For all of these reasons, we contend that interrater reliability is best treated as a descriptive statistic that describes properties of a specific set of data that were obtained from a specific use of the scale.

3. Computation of R

Operationally, the decision about whether to include the between-judges variance in the error term involves the choice between computing a one-way analysis of variance (ANOVA) or a two-way ANOVA (Ebel, 1951; Guilford, 1954). Use the two-way procedure when you do not want to include the between-judges variance in the error term (i.e., when you plan to make a norm-referenced interpretation about the ordering of the rated objects relative to each other). Given the assumption of no interaction between persons and judges, the analysis involves the use of standard ANOVA procedures to compute the mean square for persons (MS_p), mean square for judges (MS_j), mean square for error (MS_e), and total mean square (MS_t). These components are then inserted into the standard equation for the intraclass correlation:

$$R_i = \frac{MS_p - MS_e}{MS_p + MS_e(K - 1)} \quad (1)$$

where i indicates that the average reliability of a single (i.e., individual) rater is estimated, a point we will explain in the next section, and K = the number of judges ratings each person. Use of this approach to calculate the interrater reliability for Cases 1, 2, and 3 in Table 4.1 yields $R_{di} = 1.00$, 1.00, and 0.38, respectively (see Table 4.1), where d signifies that the reliability is descriptive of the data on which it is calculated.

Suen and Ary (1989) suggested that R_{di} be interpreted as a measure of interrater agreement, but Case 2 illustrates the problem with that approach (see Table 4.1). Interpreting R_{di} as a measure of agreement leads to the conclusion that the Case 2 raters were in perfect agreement (i.e., $R_{di} = 1.00$) when in fact the raters did not agree on the rating to be assigned to a single job candidate. Instead, R_{di} indicates the extent to which the judges ranked the objects in the same order.

You may wonder why the MS_j was computed but then not used in Equation 1 to compute R . In ANOVA the error term is the residual that remains after all of the desired components have been subtracted from the MS_t . Computationally, therefore, calculation of the MS_j component in the two-way ANOVA is tantamount to removing it from the error term.

The easiest way to compute the components for Equation 1 when the between-judges variance is to be included in the error term is to use the standard one-way ANOVA procedure, in which only mean square for persons MS_p and mean square of error MS_e are calculated (Ebel, 1951; Tinsley & Weiss, 1975). The one-way ANOVA procedure does not extract a variance component to estimate the between-judges variability, thereby leaving the mean square for judges in the residual that is used as an estimate of error. Following the computation of its components, the interrater reliability is calculated using Equation 1. When the between-judges variance is interpreted as error, the intraclass correlations for Cases 1, 2, and 3 in Table 4.1 are $R_{gi} = 1.00, .017$, and $.40$, respectively (see Table 4.1), where $_g$ indicates that the intraclass correlation is generalizable to the population of judges from which the sample of judges was drawn, and $_i$ indicates that the average reliability of a single (i.e., individual) rater is estimated. The difference between the two estimates of reliability is greatest for Case 2 (see Table 4.1), where substantial differences were observed in the means of the raters.

Frequently, applied researchers are confronted with incomplete data. When the objects have been rated by varying numbers of raters and the investigator plans to use the one-way ANOVA procedure, the following equation from Snedecor (1946) yields an average value of K that can be used in Equation 1:

$$\bar{K} = \frac{1}{N-1} \left(\sum K - \frac{\sum K^2}{\sum K} \right), \quad (2)$$

where $\bar{K}$ = the average value of K to be inserted in Equation 1 for K , N = the number of subjects, and K = the number of judges rating each subject (this value will vary from subject to subject).

B. Reliability of Composite Ratings

When measuring a construct using a multiple-item scale, researchers typically obtain a composite score that reflects the respondents' scores on all of the items (i.e., a mean item score or sum of the item scores). Generally, they are concerned about the reliability of this composite because judgments about the relative ordering of the respondents are based on the composite. In an analogous manner, we have recommended that researchers use the mean rating or sum of the ratings of the panel of judges for making decisions.

Therefore, it is the reliability of the composite rating that is of interest, but the value of R yielded by Equation 1 actually indicates the average reliability of the single judge. As such, Equation 1 underestimates the interrater reliability of the composite rating.

Researchers should use a measure of reliability that is consistent with the way in which the ratings data are interpreted. If conclusions typically are drawn from the ratings of a single judge, the average reliability of the individual judge (i.e., R_{di} or R_{gi}) is appropriate even if the ratings of several judges are available because decisions are based on the ratings of a single judge. Conversely, when decisions are based on a composite formed by combining information from a group of observers, it is necessary to know the reliability of the composite. This reliability will be higher than the reliability yielded by Equation 1; it can be estimated by applying the Spearman-Brown Prophecy formula (see Crocker & Algina, 1986, p. 119) to the result obtained from Equation 1, but it is quicker to calculate it directly using the following equation:

$$R_c = \frac{MS_p - MS_e}{MS_p}, \quad (3)$$

where c signifies that the intraclass correlation estimates the reliability of the composite rating.

In a manner analogous to adding items to a scale to increase its reliability, increasing the number of raters yields gains in interrater reliability when the initial reliability is low (Guilford, 1954). Interrater reliability will increase rapidly as a result of adding the first several raters, but decreasing increments in reliability occur with the addition of each new rater. Use of the variance components derived from the two-way ANOVA in Equation 3 yields $R_{dc} = 1.00, 1.00$, and $.65$ for Cases 1, 2, and 3 in Table 4.1, respectively, where d indicates the interrater reliability is descriptive of the sample on which it is calculated. Use of the variance components derived from the one-way ANOVA yields $R_{gc} = 1.00, .06$, and $.67$ for Cases 1, 2, and 3 in Table 4.1, respectively, where g signifies that the reliability is generalizable to other situations, given the restrictive conditions specified previously.

1. Differential Weighting

Equation 3 and the Spearman-Brown formula assign unit weights to the ratings of each judge (i.e., weights them equally) in estimating the reliability of the composite rating. This is the same as the procedure used in calculating a composite score from the responses to a multiple-item scale (i.e., you add the score for item one to the score for item two, and so on). In this instance, however, you might wonder whether differentially weighting the ratings based on some a priori information about the quality of the judges would yield a higher estimate of interrater reliability than

Equation 3. For example, Kelley (1947) suggested calculating the pairwise correlation among the judges and assigning weights to the ratings of the different observers based on their estimated reliability. When calculating a multiple-item composite score, the unit weighting scheme (i.e., simply adding the item scores) actually weights the items according to their item variances. That is because the variance of a composite is equal to the sum of the variances of the items forming the composite plus the sum of their interitem covariances. The same general principle holds when ratings are combined to produce a composite rating; the ratings are differentially weighted by the variances of the individual judges. In the extreme case, the ratings of a judge who rated every object the same would receive zero weight in a composite score.

We recommend the use of unit weights (i.e., Equation 3) in most situations. The use of differential weights can actually decrease the reliability of the composite rating under some circumstances (see Lawshe & Nagle, 1952; Overall, 1965). Furthermore, a set of optimal weights is applicable only to a specific set of observers. Any modification of the rating scale, change in the membership of the group of judges, or change in the reliability or variance of the judges' ratings will require the calculation of a new set of optimal weights. For example, in many rating situations judges receive consistent feedback about the reliability of their ratings. This information may well be used by the low-reliability raters to improve their reliability. For this reason, optimal weights cannot be assumed to be stable across rating occasions, even if the same group of judges uses the same rating scale in rating the same or a similar group of objects.

C. Reliability When Variance Is Low

We previously noted that R can be greatly reduced when the variance in the ratings is low (see Case 3). The question remains about what to do in such situations. Rather than immediately discard the intraclass correlation for another statistic, we recommend you first determine the cause of the low variance. It may be that the judges are using the rating scale improperly. When that is the case it is best to provide additional training until the judges' performance meets the standard you have established for an acceptable level of proficiency. Another possibility is that a homogenous sample of objects is being rated. When that occurs and the sample of objects is not representative of the population, the best strategy is to obtain a more representative sample for rating. However, it is possible in applied research that the judges are performing competently and the sample, although rather homogeneous, is representative of the population or is of interest in its own right. In those circumstances it may be necessary to use an alternative estimate of interrater reliability.

Finn (1970) developed the following index of interrater reliability for use with ordinal-scaled ratings when within-judge variance is low:

$$r_i = 1.0 - \frac{MS_e}{(K^2 - 1)/12}, \quad (4)$$

where MS_e = mean square for error obtained from a one-way ANOVA (unless only one object has been rated, in which cases MS_e = the variance of the assigned ratings), K = the number of scale categories, and i indicates that the average reliability of a single (i.e., individual) rater is estimated. Computationally, the denominator is the variance expected if the ratings were assigned at random; conceptually, it provides an estimate of the total variance. Therefore, the ratio indicates the proportion of total variance that is attributable to error variance and r_i indicates the proportion of total variance that is attributable to true score variance (i.e., to genuine differences among the rated objects on the attribute being rated). An r_i of 1.00 indicates perfect reliability; an r_i of 0 indicates that the observed ratings varied as much as chance ratings. For the data in Table 4.1, $r_i = 1.00$, .23, and .94 for Cases 1, 2, and 3, respectively. Comparison of Finn's r_i with the four methods of computing R is informative. The results for Case 3 reveal that r_i is, indeed, relatively independent of the variance in the ratings, whereas Case 2 illustrates that r_i treats between-judges variance as error.

Finn's r_i , like R_{gi} , indicates the average reliability of a single rater. Application of the Spearman-Brown formula to the values obtained from Equation 4 reveals that the reliability of the composite rating (i.e., r_c) for Cases 1–3 in Table 4.1 is 1.00, .47, and .98, respectively.

Finn's r_i is a descriptive statistic; in actual practice the random assignment of ratings by a set of observers could result in an observed variance that is smaller than the expected variance purely by chance. Researchers can test the hypothesis that the observed variance is equal to the chance variance using the following chi-square test:

$$\chi^2 = \frac{N(k - 1)MS_e}{(K^2 - 1)/12}, \quad (5)$$

where N = number of subjects, $N(k - 1)$ = the degrees of freedom, and MS_e and K are defined in Equation 4. A chi-square value lower than the lower critical value for the one-tailed chi-square test (i.e., the .99 level for a test at $p \leq .01$) would indicate that the observed variance is significantly less than the chance variance. The interrater reliability is significantly greater than zero only if the null hypothesis is rejected.

We recommend a stringent critical value (e.g., $p < .01$) because this use of chi-square assumes that the ratings will be normally distributed and that both the subjects and raters will be randomly selected. Typically, these assumptions will be violated in applied research. Violation of the normality

assumption are quite serious when inferences are made about variances (Hays, 1994, p. 359). Also keep in mind that a significant chi-square indicates only that the pattern of ratings is not consistent with the hypothesis of completely random responding. Finally, we caution that computation of the chance variance requires the assumption that the ratings of the judges are purely random and that every rating has the same probability. This assumption is violated whenever the judges have a response set to avoid the extreme categories of the rating scale. Whenever that occurs, the estimated chance variance used in Equation 4 will exceed the "true" chance variance, thereby causing r_i (and r_c) to be spuriously high.

V. INTERRATER AGREEMENT

Berk (1979) identified 16 different interrater agreement indices and numerous indices have been proposed since that time (e.g., James et al., 1984; Lindell & Brandt, 1997; Szalai, 1998). These measures include simple approaches such as calculating the pairwise correlation between observers ratings, approaches based on ANOVA and information theory (K. Lu, 1971), a variety of chi-square indices, occurrence-nonoccurrence indices (e.g., Hawkins & Dotson, 1975), and the proportion or percentage of agreement (p).

Many of the errors in assessing interrater agreement are attributable to the use of statistical indexes designed for other purposes as measures of interrater agreement. For example, pairwise correlations would be better suited for use as a measure of interrater reliability than as a measure of interrater agreement. Numerous chi-square indices have been used as measures of interrater agreement, but it is not possible to draw any inference about the degree of agreement from a chi-square test alone. As we noted previously (Tinsley & Weiss, 1975), chi-square tests the hypothesis that the proportion of objects assigned to the various rating categories by the different judges do not differ significantly. A nonsignificant chi-square merely indicates that the observed disagreement is not greater than the disagreement that could be expected on the basis of chance, whereas a significant chi-square indicates that the amount of agreement is greater than or less than expected on the basis of chance.

One deceptively simple statistic is the percentage (or proportion) of agreements, defined as

$$\rho = \frac{N_a}{N_a + N_d}, \quad (6)$$

where N_a = the number of agreements and N_d = the number of disagreements. $\rho = 1.00$, $.00$ and $.30$, respectively, for the three cases in Table 4.1. Despite repeated criticisms (e.g., Cohen, 1960; Kelly, 1977; Mitchell, 1979;

Robinson, 1957; Tinsley & Weiss, 1975), variations of ρ continue to be the most widely used measures of interrater agreement. This is most likely attributable to its ease of calculation and its misleadingly simple interpretation.

Several problems limit the usefulness of ρ as an index of interrater agreement. First, ρ treats interrater agreement in an absolute, all-or-none fashion. For example, $\rho = 0$ for the data in Case 2, despite the fact that the judges were in perfect agreement on the relative merits of the job applicants. Another problem is that some agreements among observers can be expected on the basis of chance alone. For example, if two judges made a series of dichotomous ratings at random, their ratings would be expected to agree 25% of the time purely on the basis of chance. ρ ignores the likelihood of chance agreements and thereby overestimates the true agreement by an amount related to the number of raters and number of points on the scale. Suen and Lee (1985) reanalyzed data from a sample of published studies and concluded that correcting for chance agreements would have led to the conclusion that the quality of the ratings data was unacceptable in 25% to 75% of these studies. Nevertheless, these difficulties have been partially avoided in the better measures of interrater agreement, and ρ is the foundation upon which these measures of agreement are constructed.

A. Nominal Scales

Numerous writers have suggested measures of interrater agreement for nominal scales based on the percentage or proportion of agreements among judges (p) (Cohen, 1960, 1968; Everitt, 1968; Fleiss, 1971; Fleiss, Cohen, & Everitt, 1969; Goodman & Kruskal, 1954; Scott, 1955). Although p treats agreement as an absolute, that is not a disadvantage if all of the disagreements are regarded as equally serious, as is typically the case when dealing with nominal ratings. Nevertheless, the problem of chance agreement remains. Guttman, Spector, Sigal, Rakoff, and Epstein (1971) concluded that there was "tacit" consensus that 65% agreement represented the minimum acceptable standard, but fortunately more discriminating measures are available that represent observed agreement as a function of chance agreement and provide statistical tests of the significance of ρ .

1. Two Judges

Cohen's (1960) coefficient kappa (κ) indicates the proportion of agreements between two raters after adjusting for chance agreements. κ is calculated as follows:

$$\kappa = \frac{P_o - P_c}{1 - P_c}, \quad (7)$$

where P_o = the proportion of ratings in which the two judges agree, and P_c = the proportion of ratings for which agreement is expected by chance. The expected chance agreement can be obtained as follows:

$$P_c = \sum_{j=1}^k \hat{P}_{jj}, \quad (8)$$

where $\hat{P}_{jj}$ = the product of the marginal proportions. For the data in Table 4.2, which shows two judges' hypothetical classifications of 100 interview statements, $P_c = [(.20 \times .30) + (.30 \times .20) + (.30 \times .40) + (.20 \times .10)] = .26$, $P_o = [.18 + .18 + .24 + .10] = .70$, and κ is .59.

κ can vary from 1.00 (which indicates perfect agreement among the judges) to -1.00 when the marginal frequencies are identical. A κ value of 0 indicates that the observed agreement is equal to the agreement that would be expected by chance, whereas a negative value indicates that the observed agreement is less than the expected chance agreement.

Several measures of interrater agreement have been suggested that are identical in form to κ but differ in their definition of chance agreement (e.g., Goodman & Kruskal, 1954; Scotts, 1955), but each makes more restrictive assumptions than κ (e.g., the assumption of equal distribution of ratings). In contrast, κ recognizes that judges distribute their ratings differently across the categories and the only assumptions required are that (a) the objects to be rated are independent, (b) the judges assign their ratings independently, and (c) the categories of the nominal scale are independent, mutually exclusive, and exhaustive. Formulas for the standard error of κ , for testing the significance of the difference between two kappa coefficients, and for testing the hypothesis that $\kappa = 0$ are provided by Cohen (1960) and Fleiss et al. (1969). Fleiss (1971) has formulated an extension of kappa (κ_v) for measuring interrater agreement when subjects are rated by different sets of judges, but the number of judges per subject is constant, and for evaluating the interrater agreement regarding assignments to a particular rating category.

TABLE 4.2 Proportion of 100 Interview Statements Classified by Two Reviewers as Indicative of Four Psychological Motivations

Reviewer B	Reviewer A				Total
	Autonomy	Affiliation	Competence	Status	
Autonomy	.18	.00	.02	.00	.20
Affiliation	.00	.18	.12	.00	.30
Competence	.06	.00	.24	.00	.30
Status	.06	.02	.02	.10	.20
Total	.30	.20	.40	.10	—

2. Weighted Kappa

κ is based on the assumption that all disagreements in classification are equally serious, but in some instances researchers may consider some disagreements among judges to be more serious than others. For example, if two therapists were diagnosing patients as normal, neurotic, psychotic, or suicidal, disagreement as to whether a patient was normal as opposed to psychotic or suicidal might be regarded as more serious than disagreement over whether the patient was normal or neurotic. Cohen (1968) developed weighted kappa (κ_w) for use as an index of interrater agreement when researchers want to differentially weight the various disagreements that occur in assigning objects to nominal scale categories. κ is actually a special case of κ_w in which all agreements (i.e., the diagonal entries in Table 4.2) are assigned weights of 1.0 and all disagreements (the off-diagonal values in Table 4.2) are assigned weights of 0. However, κ_w permits the assignment of variable weights to the various types of disagreements (i.e., the off-diagonal cells). Cohen (1968), Everitt (1968), and Fleiss et al. (1969) have provided computational formulas and examples.

Although this option may sound attractive, our prediction that there would be few valid uses of κ_w has proven to be accurate (Tinsley & Weiss, 1975). Assigning weights to order nominal rating categories along a unidimensional continuum transforms nominal ratings to ratings possessing the properties of ratio measurement. Under these circumstances, κ_w is identical to the intraclass correlation (Fleiss & Cohen, 1973), and κ_w actually measures interrater reliability, not interrater agreement, but it lacks the flexibility and ease of computation of the intraclass correlation. The other possibility is to weight the nominal rating categories using some nonlinear (i.e., multi-dimensional) scheme. That is most likely to occur when the researcher is operating on the basis of an explicit theory about the relations among the rating categories. When κ_w is used the investigator must specify the weights used in the analysis and explain in detail the basis upon which the weights were assigned.

B. Ordinal and Interval Scales

A central tenet of decision theory (Cronbach & Gleser, 1965) is that errors of measurement differ greatly in their importance. Some very small errors can be quite serious (e.g., a mistake of one gram in preparing a prescription or a mistake of one degree in computing the trajectory of the space shuttle), whereas in other instances quite large differences are of no appreciable consequences (e.g., a mistake of one hundred pounds when measuring 100 metric tons of grain or a mistake of one mile when estimating the distance from New York to Los Angeles). This essential truth is often ignored in the sciences and humanities, where a premium is placed on achieving the greatest precision possible. For example, measures of interrater reliability,

by their nature, emphasize precision, and departures from the linear model underlying these measures are interpreted as serious deficiencies in the ratings. Nevertheless, failure to distinguish between important and inconsequential errors can be limiting and even self-defeating. The assessment of interrater agreement differs from the assessment of interrater reliability in this critical aspect. It is not possible to assess interrater agreement adequately without considering the seriousness or cost of the various types of disagreements that can occur.

Lawlis and Lu (1972) developed a measure of interrater agreement that allows the researcher the option of defining agreement as identical ratings, as ratings that differ by no more than 1 point, or as ratings that differ by no more than 2 points. Their flexible model allows researchers to judge the seriousness of disagreements among the raters and to distinguish between serious and unimportant disagreements. Agreement is tallied one object at a time, by determining whether the total set of ratings given that object satisfies the criterion. For example, three interviewers from a multinational corporation who gave a job applicant ratings of 1, 2, and 3 on the scale illustrated in Table 4.1 would have disagreed in an absolute sense, but essentially all would have rated the applicant as low in the qualities viewed as desirable for employment. If agreement were defined as ratings that differed by not more than two scale categories, the Lawlis and Lu index of agreement would score the set of ratings as an agreement.

Lawlis and Lu (1972) proposed a nonparametric chi-square test of the significance of interrater agreement, but neglected to provide a descriptive index of agreement. Tinsley and Weiss (1975) proposed the *T*-index, patterned after Cohen's (1960) κ , as a measure of agreement:

$$T = \frac{N_a - N\rho_c}{N - N\rho_c}, \quad (9)$$

where N_a = the number of agreements, N = the number of individuals rated, and ρ_c = the probability of chance agreement on an individual. The Appendix indicates the probability of chance agreement when agreement is defined as 0, 1, and 2 points discrepancy.

Positive values of T indicate that the observed agreement is greater than chance agreement, negative values indicate that the observed agreement is less than chance agreement, and T is zero when the observed agreement is equal to the expected chance agreement. The ratings for Case 3 of Table 4.1 can be used to illustrate the flexibility of this approach and the critical importance of the operational definition of agreement adopted. When agreement is defined as 0, 1, and 2 points discrepancy, respectively, the number of agreements equals 1 (applicant J), 7 (all applicants except C, D, and I), and 10 (all of the applicants), and the corresponding T values are $T_0 = .00$, $T_1 = .67$, and $T_2 = 1.00$.

The statistical significance of T can be tested using Lawlis and Lu's (1972) nonparametric chi-square, with 1 degree of freedom:

$$\chi^2 = \frac{(N_a - N\rho_c - .5)^2}{N\rho_c} + \frac{(N_d - N(1 - \rho_c) - .5)^2}{N(1 - \rho_c)}, \quad (10)$$

where N_a , N , and ρ_c are defined as in Equation 9, $.5$ = a correction of continuity, and N_d = the number of disagreements. A significant chi-square indicates that the observed agreement is greater than the agreement that could be expected on the basis of chance, whereas a nonsignificant chi-square means that the hypothesis that the ratings were assigned at random cannot be rejected. Several factors could be responsible for the failure to find significant agreement, including characteristics of the scale (e.g., vaguely defined or overlapping categories), the judges (e.g., carelessness or lack of adequate training), or the objects (e.g., homogeneity of the sample or inconsistent expression of the attribute being rated). Continued use of the scale in the manner in which the data were obtained is ill advised when a nonsignificant chi-square is obtained.

Researchers must study their rating scales and think carefully about the implications of the alternative definitions of agreement for their research, for the T -index and the results of Lawlis and Lu's (1972) chi-square are specific to the definition of agreement. The advantages of being able to distinguish disagreements that have no serious consequences from those that are of practical significance carries with it the temptation to manipulate the definition of agreement so that the results will reach the desired level of agreement. Consequently, the definition of agreement should be determined on theoretical, empirical, or practical grounds before data analysis has begun, and preferably before data collection has begun. Furthermore, the definition of agreement and the rationale for its adoption should be stated when reporting the interrater agreement of the ratings.

One further consideration is that Lawlis and Lu's (1972) estimate of chance agreement is based on the assumption that every judgment has the same probability. We pointed out earlier that under some circumstances (e.g., when raters do not use the end categories of a rating scale), the true probability of chance agreement (ρ_c) is greater than $N\rho_c$. In practice, therefore, $N\rho_c$ is the lower limit for the unknown probability of chance agreement, (ρ_c) is often underestimated, and the significance of the observed agreement is overstated. Careful investigators can avoid underestimating (ρ_c) by adjusting the required level of significance from the traditional .05 to a more stringent level, or by calculating the probability of chance agreement on fewer scale categories than are actually available (e.g., use the probability of chance agreement for a 5- or 6-point scale when a 7-point scale was used).

VI. SUMMARY OF RECOMMENDATIONS

Evidence regarding both interrater reliability and interrater agreement should be reported whenever rating scales are used. Interrater reliability and agreement are functions of the objects rated, the scale used, and the observers making the ratings, so they are descriptive of a specific set of data. Attempts to generalize interrater reliability and agreement from other research or other groups of raters is misleading. All of the recommended measures of interrater reliability and agreement are one-trial estimates, so calculation of these indices should pose no logistic problems if the study has been designed properly.

The intraclass correlation (Equation 1) is the best measure of interrater reliability for ordinal and interval level measurement; nominally scaled data permit an analysis only of interrater agreement. Use Finn's r_c (Equation 4) only when the within-raters variance is so severely restricted that the intraclass correlation is inappropriate (e.g., Case 3 in Table 4.1). Interpret Finn's r_c cautiously because chance variance may be overestimated, thereby spuriously inflating r_c . Be explicit in explaining whether between-raters' variance was included or excluded from the error term, the implications of that decision for the possibility of generalizing your results, and whether the reported reliability represents the average reliability of a single rater or the reliability of the composite rating.

The reliability of composite scores based on unit weights will be satisfactory for most situations. Use optimal weights only when the decision to be made requires the greatest possible measurement precision and the ratings of the judges differ in variance or reliability. The procedure developed by Overall (1965) is appropriate in this situation, but take care to report the variance and estimated reliability of each judge, the weights assigned the ratings of each judge, and the estimated reliability of the composite based on unit weights and optimal weights. Also keep in mind that the optimal weights were tailored to your unique rating situation and cannot be generalized across rating occasions, even if the same judges used the same rating scale with the same or a similar sample of objects.

Use the T -Index (Equation 9) and Lawlis and Lu's chi-square (Equation 10) to estimate the interrater agreement of ordinal and interval scaled ratings. Use a stringent level of significance in evaluating agreement or a value of ρ_c based on fewer scale categories than were actually available to correct for the possibility that the raters were predisposed to avoid the extreme categories of the scale. Report all decisions made in conducting the analysis, including the definition of agreement adopted and the rationale for the definition.

With nominal scaled ratings use Cohen's κ (Equation 7) when the same two judges rate each object and Fleiss's κ_v when the objects are rated by

different judges but the same number of judges rate each object. We do not recommend the use of weighted kappa (κ_w).

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APPENDIX Probability of Chance Agreement for Three Definitions of Agreement^a

Points on scale	Number of judges												
	2	3	4	5	6	7	8	9	10	11	12	13	14
Agreement defined as zero points discrepancy in ratings													
2	.500	.250	.125	.063	.031	.106	.008	.004	.002	.001	.001	—	—
3	.333	.111	.037	.012	.004	.001	.001	—	—	—	—	—	—
4	.250	.063	.016	.004	.001	—	—	—	—	—	—	—	—
5	.200	.040	.000	.002	—	—	—	—	—	—	—	—	—
6	.167	.028	.005	.001	—	—	—	—	—	—	—	—	—
7	.143	.020	.003	—	—	—	—	—	—	—	—	—	—
8	.125	.016	.002	—	—	—	—	—	—	—	—	—	—
9	.111	.012	.001	—	—	—	—	—	—	—	—	—	—
10	.100	.010	.001	—	—	—	—	—	—	—	—	—	—
11	.091	.003	.001	—	—	—	—	—	—	—	—	—	—
12	.083	.007	.001	—	—	—	—	—	—	—	—	—	—
13	.077	.006	.001	—	—	—	—	—	—	—	—	—	—
14	.071	.005	—	—	—	—	—	—	—	—	—	—	—
15	.067	.004	—	—	—	—	—	—	—	—	—	—	—
16	.063	.004	—	—	—	—	—	—	—	—	—	—	—
17	.059	.004	—	—	—	—	—	—	—	—	—	—	—
18	.056	.003	—	—	—	—	—	—	—	—	—	—	—
19	.053	.003	—	—	—	—	—	—	—	—	—	—	—
20	.050	.003	—	—	—	—	—	—	—	—	—	—	—
21	.048	.002	—	—	—	—	—	—	—	—	—	—	—
22	.046	.002	—	—	—	—	—	—	—	—	—	—	—
23	.044	.002	—	—	—	—	—	—	—	—	—	—	—
24	.042	.002	—	—	—	—	—	—	—	—	—	—	—
25	.040	.002	—	—	—	—	—	—	—	—	—	—	—
26	.039	.002	—	—	—	—	—	—	—	—	—	—	—
27	.037	.001	—	—	—	—	—	—	—	—	—	—	—
28	.036	.001	—	—	—	—	—	—	—	—	—	—	—
29	.035	.001	—	—	—	—	—	—	—	—	—	—	—
30	.033	.001	—	—	—	—	—	—	—	—	—	—	—
31	.032	.001	—	—	—	—	—	—	—	—	—	—	—
32	.031	.001	—	—	—	—	—	—	—	—	—	—	—
33	.030	.001	—	—	—	—	—	—	—	—	—	—	—
34	.029	.001	—	—	—	—	—	—	—	—	—	—	—
35	.029	.001	—	—	—	—	—	—	—	—	—	—	—

(continued)

APPENDIX (continued)

Points on scale	Number of judges												
	2	3	4	5	6	7	8	9	10	11	12	13	14
Agreement defined as $\leq$ one point discrepancy in ratings ^b													
3	.778	.556	.383	.259	.174	.117	.078	.052	.035	.023	.015	.010	.007
4	.625	.344	.180	.092	.046	.023	.012	.006	.003	.002	.001	—	—
5	.520	.232	.098	.040	.0162	.007	.003	.001	—	—	—	—	—
6	.444	.167	.059	.020	.007	.002	.001	—					
7	.388	.125	.038	.011	.003	.001	—	—					
8	.343	.098	.026	.007	.002	—	—	—					
9	.309	.078	.018	.004	.001	—	—	—					
10	.280	.064	.014	.003	.001	—	—	—					
11	.256	.053	.010	.002	—								
12	.236	.045	.008	.004	—								
13	.219	.039	.006	.001	—								
14	.204	.034	.005	.001	—								
15	.191	.029	.004	.001	—								
16	.180	.026	.003	—									
17	.170	.023	.003	—									
18	.161	.021	.002	—									
19	.152	.019	.002	—									
20	.145	.017	.002	—									
21	.138	.015	.002	—									
22	.132	.014	.001	—									
23	.127	.013	.001	—									
24	.122	.012	.001	—									
25	.117	.011	.001	—									
26	.112	.010	.001	—									
27	.108	.009	.001	—									
28	.105	.009	.001	—									
29	.101	.008	.001	—									
30	.098	.008	.001	—									
31	.095	.007	.001	—									
32	.092	.007	—	—									
33	.089	.006	—	—									
34	.087	.006	—	—									
35	.084	.006	—	—									

(continues)

APPENDIX (continued)

Points on scale	Number of judges												
	2	3	4	5	6	7	8	9	10	11	12	13	14
Agreement defined as $\leq$ two points discrepancy in ratings ^c													
4	.875	.719	.570	.443	.340	.259	.196	.148	.117	.084	.063	.047	.036
5	.760	.520	.338	.213	.132	.081	.049	.030	.02	.11	.007	.004	.002
6	.667	.389	.213	.113	.059	.0300	.015	.008	.004	.002	.001	.001	—
7	.592	.300	.142	.065	.029	.013	.006	.002	.001	—	—	—	—
8	.531	.238	.099	.040	.016	.006	.002	.001	—	—	—	—	—
9	.482	.193	.072	.026	.009	.003	.001	—	—	—	—	—	—
10	.440	.160	.054	.017	.005	.002	.001	—	—	—	—	—	—
11	.405	.135	.041	.012	.003	.001	—	—	—	—	—	—	—
12	.375	.115	.032	.009	.002	.001	—	—	—	—	—	—	—
13	.349	.099	.026	.006	.002	—	—	—	—	—	—	—	—
14	.327	.086	.021	.005	.001	—	—	—	—	—	—	—	—
15	.307	.076	.017	.004	.001	—	—	—	—	—	—	—	—
16	.289	.067	.014	.003	.001	—	—	—	—	—	—	—	—
17	.273	.060	.012	.002	—	—	—	—	—	—	—	—	—
18	.259	.054	.010	.002	—	—	—	—	—	—	—	—	—
19	.247	.048	.009	.002	—	—	—	—	—	—	—	—	—
20	.235	.044	.008	.001	—	—	—	—	—	—	—	—	—
21	.225	.040	.006	.001	—	—	—	—	—	—	—	—	—
22	.215	.037	.006	.001	—	—	—	—	—	—	—	—	—
23	.206	.034	.005	.001	—	—	—	—	—	—	—	—	—
24	.198	.031	.004	.001	—	—	—	—	—	—	—	—	—
25	.190	.029	.004	.001	—	—	—	—	—	—	—	—	—
26	.183	.026	.003	—	—	—	—	—	—	—	—	—	—
27	.177	.025	.003	—	—	—	—	—	—	—	—	—	—
28	.171	.023	.003	—	—	—	—	—	—	—	—	—	—
29	.165	.021	.003	—	—	—	—	—	—	—	—	—	—
30	.160	.020	.002	—	—	—	—	—	—	—	—	—	—
31	.155	.019	.002	—	—	—	—	—	—	—	—	—	—
32	.150	.018	.002	—	—	—	—	—	—	—	—	—	—
33	.146	.017	.002	—	—	—	—	—	—	—	—	—	—
34	.142	.016	.002	—	—	—	—	—	—	—	—	—	—
35	.138	.015	.001	—	—	—	—	—	—	—	—	—	—

(continues)

APPENDIX (continued)

Points on scale	Number of judges									
	Zero Points			$\leq$ One Point			$\leq$ Two Points			
	2	3	4	2	3	4	2	3	4	5
36	.028	.001	—	.082	.005	—	.134	.014	.001	—
37	.027	.001	—	.080	.005	—	.131	.013	.001	—
38	.026	.001	—	.078	.005	—	.127	.013	.001	—
39	.026	.001	—	.076	.005	—	.124	.012	.001	—
40	.025	.001	—	.074	.004	—	.121	.011	.001	—
41	.024	.001	—	.072	.004	—	.118	.011	.001	—
42	.024	.001	—	.070	.004	—	.006	.010	.001	—
43	.023	.001	—	.069	.004	—	.113	.010	.001	—
44	.023	.001	—	.067	.004	—	.111	.010	.001	—
45	.022	.001	—	.066	.003	—	.108	.010	.001	—
46	.022	.001	—	.064	.003	—	.106	.009	.001	—
47	.021	.001	—	.063	.003	—	.104	.008	.001	—
48	.021	—	—	.062	.003	—	.102	.008	.001	—
49	.020	—	—	.060	.003	—	.010	.008	.001	—
50	.020	—	—	.059	.003	—	.010	.007	.001	—

^a Tables computed from formulas appearing in Lawlis and Lu (1972).

^b Probability of chance agreement for a 3-point scale when agreement defined as ≤ 1 point discrepancy equals .005 (15 judges), .003 (16 judges), .002 (17 judges), and .001 (≥ 18 judges).

^c Probability of chance agreement for a 4-point scale when agreement defined as ≤ 2 points discrepancy equals .027 (15 judges), .020 (16 judges), .015 (17 judges), .011 (18 judges), .009 (19 judges), .006 (20 judges), .005 (21 judges), .004 (22 judges), .003 (23 judges), .002 (24 judges), .002 (25 judges), and .001 (≥ 26 judges). Probability of chance agreement for a 5-point scale when agreement defined as ≤ 2 points discrepancy equals .001 (≥ 15 judges).